Enhancing Underwater Debris Detection Using Global–Local Networks and YOLOv12

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***Abstract*— Marine debris, particularly plastic waste, poses a persistent threat to underwater ecosystems due to its longevity and widespread distribution. This paper introduces a deep learning framework for underwater debris detection that integrates global–local image enhancement with YOLOv12 for real-time object detection. The enhancement module consists of a Global branch for correcting color distortions and a Local branch, based on Feature Pyramid Network (FPN) and Squeeze-and-Excitation (SE), for improving spatial contrast. Experimental results demonstrate strong performance, achieving an average Precision of 93.22%, Recall of 90.58%, and mean Average Precision (mAP) of 93.14% across 16 object categories, with several debris types exceeding 96% mAP. Beyond technical accuracy, this work highlights the potential of the framework in terms of AI competitiveness and Big Data integration. Its real-time capability enables deployment on autonomous underwater vehicles (AUVs) and robotic platforms, while its modular design facilitates cloud-based scalability for large-scale ocean monitoring. Furthermore, the system can be extended to multi-modal environmental data fusion, supporting sustainable marine management and global conservation initiatives. Thus, the contribution lies not only in performance gains but also in advancing scalable, competitive AI solutions for environmental challenges.**

Keywords*— *marine debris detection, underwater image enhancement, deep learning, YOLOv12.

I. INTRODUCTION

Marine debris has emerged as a pressing global concern, posing severe threats to marine ecosystems and biodiversity. Among the most hazardous pollutants is plastic waste, which can persist in ocean environments for centuries and cause extensive damage through ingestion, entanglement, and habitat disruption. The urgency to monitor and mitigate these impacts is increasingly recognized, yet conventional survey methods remain inadequate due to the vast scale of the ocean and the complexity of underwater environments. Challenges such as low visibility, light scattering, color distortion, and the heterogeneous appearance of submerged objects further complicate manual detection.

To address these limitations, this research introduces a deep learning-based framework specifically designed to enhance underwater image quality and improve debris detection accuracy. The system integrates multi-scale representation learning and attention-based enhancement to cope with the heterogeneity and visual distortion commonly encountered in underwater scenes. A hierarchical feature processing strategy combines fine spatial details with global semantic context, enabling the recognition of subtle debris patterns that might otherwise be overlooked. In addition, feature recalibration techniques are incorporated to amplify the most informative features while suppressing background noise, thereby improving robustness without incurring excessive computational cost. The enhancement module is guided by perceptual objectives, including SSIM and average color losses, ensuring that the generated images closely resemble high-quality references and retain critical structural cues for subsequent detection tasks.

For object detection, the framework employs YOLOv12, which offers significant advancements over prior versions. Its architecture is optimized for detecting small, irregular, and partially occluded objects—typical characteristics of marine debris. Through refined FPN and SE modules, YOLOv12 effectively extracts both high-level semantic information and fine-grained spatial details, aligning closely with the proposed Global–Local enhancement approach. These improvements enable the system to balance contextual awareness with precise object localization.

Moreover, YOLOv12 demonstrates robustness under challenging underwater conditions such as low-light, noisy, and blurry imagery. Enhanced data augmentation, a modernized backbone, and optimized training strategies strengthen generalization, while real-time inference speed is preserved making the model well-suited for deployment on autonomous underwater vehicles (AUVs) and other robotic platforms. Importantly, YOLOv12's compatibility with attention mechanisms allows seamless integration with the proposed enhancement modules, further improving the model's ability to focus on debris regions while reducing false detections caused by background noise.

Given these characteristics, the proposed integration of Global–Local enhancement and YOLOv12 provides a compelling solution to long-standing challenges in underwater debris detection, combining accuracy, robustness, and deployment feasibility for real-world marine conservation applications.

II. RELATED WORKS

Marine debris, particularly plastic waste, poses a critical threat to aquatic ecosystems, degrading biodiversity and water quality. Such debris including microplastics, abandoned fishing gear, and consumer waste persists for decades, disrupting food chains, releasing toxins, and increasing risks of entanglement or ingestion. Its long-term presence in shallow and deep-sea environments has intensified interest in automated detection and monitoring, since traditional surveys

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(e.g., diver observations, manual reports) are time-consuming and unsuitable for large-scale assessment.

Recent research highlights this urgency. Alshibli et al. [1] examined the toxicological effects of plastic waste, underscoring the need for intelligent vision systems to mitigate broader impacts such as oxygen depletion and acidification. Various deep learning-based approaches have been developed: Jalil et al. [2] compared machine learning classifiers with deep feature extractors; Xue et al. [3] proposed Shuffle-Xception for debris categorization.

YOLO-based models have shown strong results. Hipolito et al. [4] achieved accurate plastic detection with YOLOv3, while Alsuwaylimi [5] adapted YOLOv8-Seg for instance segmentation on TrashCan. Zhu et al. [6] further improved YOLOv8 with attention and feature fusion, and Sinthia et al. [7] benchmarked multiple waste categories with up to 92% accuracy. Alternative paradigms include Mask R-CNN with SIFT [8] and RTMDet improvements [9]. Complementary to detection, image enhancement methods such as white balancing and dehazing [10] or global–local histogram equalization with multi-scale fusion [11] remain crucial for underwater clarity.

Large-scale monitoring has also been explored. Yin and Cheng [12] studied macro-accumulations like the Great Pacific Garbage Patch, while Kshirsagar et al. [13] demonstrated low-cost pollution classification via color correction and texture analysis. Ali et al. [14] assessed spectral-based detection with MARIDA, highlighting the need for standardized datasets in large-scale surveillance.

Further YOLO adaptations include Shah et al. [15], who trained YOLOv8 on multi-class debris under turbid conditions; Saji et al. [16], who applied YOLOv8n for binary classification; and Wang et al. [17], who enhanced YOLOv5 with BiFPN and attention to improve detection of small objects. From an application perspective, Cheng and Tsao [18] developed a CNN-based mobile app for beach cleanups, while Chin et al. [19] deployed YOLOv5s on AUVs, achieving real-time detection above 150 FPS.

III. METHODOLOGY

To address these issues, we adopt an image enhancement strategy as a preprocessing stage prior to debris detection. The enhancement module is based on a dual-branch neural architecture inspired by the work of Bai et al. [11], which has shown effectiveness in compensating for both global and local degradations in underwater imagery. Specifically, the enhancement network comprises two synergistic components: a Global Network (G-Net) designed to correct large-scale color casts and illumination inconsistencies, and a Local Network (L-Net) focused on restoring fine-scale details and enhancing local contrast, as shown in Fig. 1.

The G-Net operates in a color space that emphasizes channel-wise dependencies, enabling it to learn global illumination priorities and correct blue-green dominance often observed in underwater scenes. In contrast, the L-Net operates on spatially localized regions, learning to recover edge sharpness, texture contrast, and object boundaries that are typically lost due to turbidity or scattering.

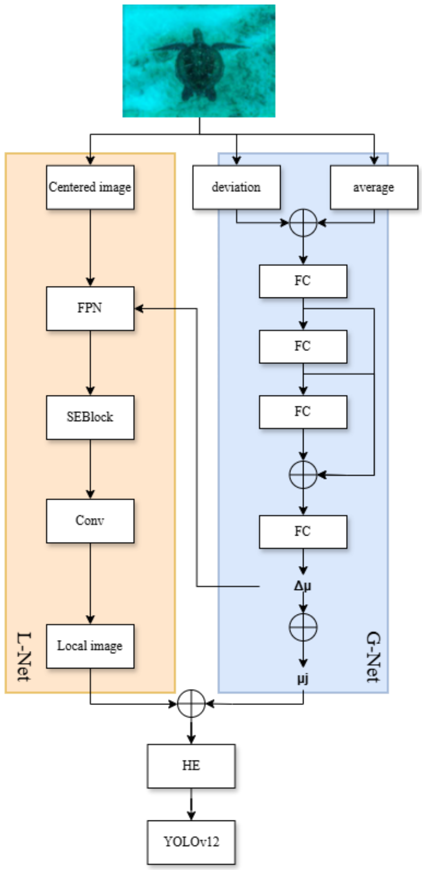

Fig. 1. Overall architecture of the proposed GlobalLocal- Feature Pyramid Network - Squeeze-and-Excitation Net for underwater image enhancement and debris detection.

The enhanced images produced by the global–local network are subsequently fed into a debris detection pipeline based on a YOLOv12 backbone, which performs object localization and classification. By decoupling the enhancement and detection stages, the framework ensures that the model benefits from visually informative input without requiring joint optimization. This modular design improves robust detection and generalization across diverse underwater environments.

A. *Global- network*

To address global color distortion caused by wavelength-dependent absorption, the Global branch (G-Net) extracts first- and second-order statistical features from the input image $I \in \mathbb{R}^C$. Specifically, the channel-wise mean $\mu_I \in \mathbb{R}^C$ and standard deviation $\sigma_I \in \mathbb{R}^C$ are computed as global features. These are concatenated into a vector:

$$x = [\mu_I, \sigma_I] \in R^{2C} \quad (1)$$

This vector is passed through a densely connected fully connected (FC) network with three hidden layers to estimate residual corrections. The updated mean is given by:

$$\hat{\mu}_J = \sigma(\mu_I + \Delta\bar{\mu}) \quad (2)$$

Where $\Delta\bar{\mu}$ is the learned residual and $\sigma(\cdot)$, denotes a sigmoid activation. This residual-learning strategy reduces the prediction burden and constrains outputs to a stable range, leading to more reliable color compensation.

B. Local Network using Feature Pyramid Network (FPN)

Although G-Net corrects global color imbalance, local contrast degradation from scattering and turbidity remains. To address this, the Local branch (L-Net) employs a Feature Pyramid Network (FPN), as illustrated in Fig. 2. The input is first normalized by subtracting the corrected global mean $\hat{\mu}_J$, ensuring that the network focuses on structural recovery rather than overall color distribution.

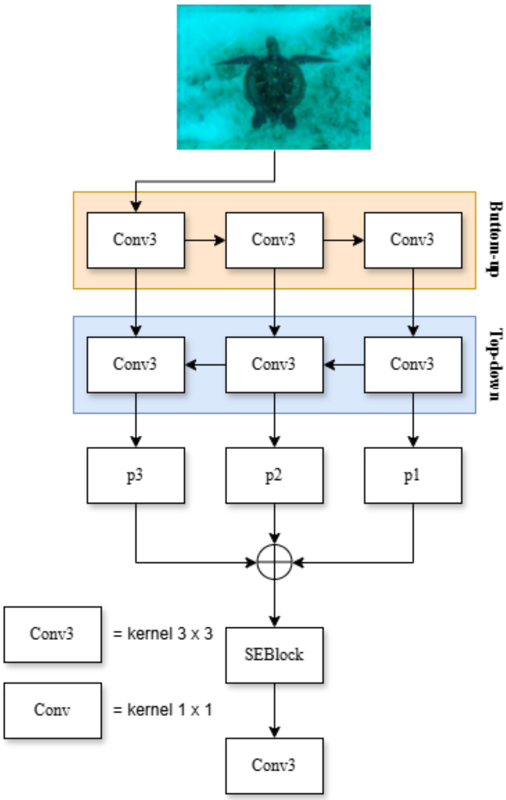

Fig. 2. Architecture of the Local Enhancement Module Based on Feature Pyramid Network (FPN).

The FPN integrates features through a bottom-up and top-down pathway with lateral connections, producing multi-scale feature maps $\{P_1, P_2, \ldots, P_n\}$. These are resized to a uniform resolution and concatenated to form a coherent multi-scale representation.

Subsequently, the fused features are passed through a Squeeze-and-Excitation (SE) block (Fig. 3), which recalibrates channel-wise importance. Global Average Pooling is applied to squeeze channel statistics, followed by fully connected layers that assign adaptive weights to emphasize channels most relevant for underwater object boundaries and fine details.

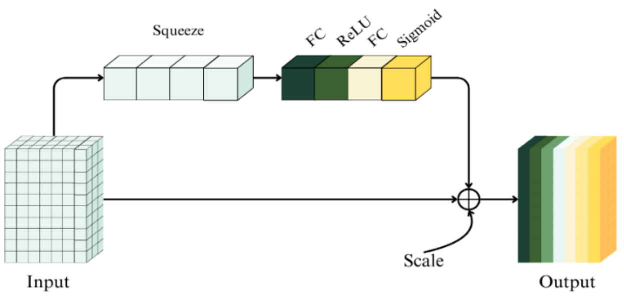

Fig. 3. Structure of the Squeeze-and-Excitation (SE) Block for Channel-wise Feature Recalibration.

This FPN–SE integration allows the L-Net to selectively enhance structural cues and local contrast, generating images with sharper edges and more balanced color tones compared to baseline global–local approaches.

C. Feature Fusion and Enhancement

To improve underwater debris detection, we propose the GlobalLocalNet, which consists of two complementary branches. The Global branch (G-Net) computes channel-wise statistics (μ_I, σ_I) and applies residual learning through fully connected layers to generate refined color means $\hat{\mu}_J$. This residual correction, constrained by a sigmoid activation, efficiently compensates wavelength-dependent color distortions.

The Local branch (L-Net) enhances fine details by processing the normalized image $I_{centered} = I - \hat{\mu}_J$ using a Feature Pyramid Network (FPN) combined with a Squeeze-and-Excitation (SE) block. As shown in Fig. 2–3, this design integrates multi-scale representations and adaptively recalibrates channel importance, allowing the model to selectively emphasize structural cues and contrast features essential for underwater scenes.

We employ two main types of loss functions: Structural Similarity Index Measure (SSIM) loss and cosine similarity-based average loss, which compares the global average color between the ground-truth and the output. SSIM loss focuses on preserving the structural and perceptual characteristics of the image, helping to retain fine details and textures in underwater imagery. The average loss, calculated using the cosine distance between the global average values of the ground-truth and the output, guides the Global branch to more accurately adjust color toward the target.

For training, we employ a hybrid loss that combines SSIM loss (to preserve textures and perceptual structure), cosine color loss (to align global color distributions), and an adversarial loss in Fig. 4. where a discriminator encourages visually natural outputs with sharper contrast. This adversarially guided enhancement improves beyond conventional SSIM-based approaches.

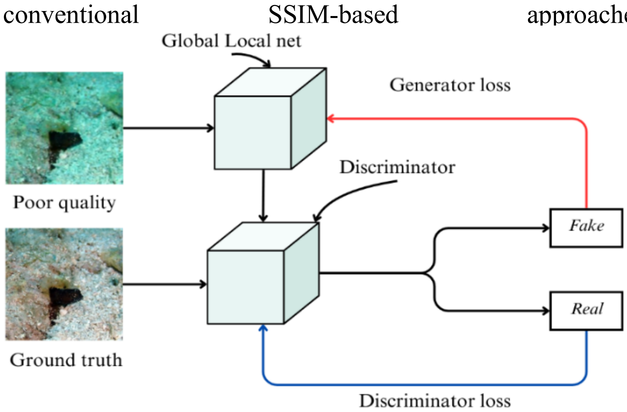

Fig. 4. Adversarial Training Framework Integrating Global–Local Network and Discriminator.

Finally, histogram equalization is applied to further refine image sharpness before detection. Enhanced images are fed into YOLOv12, which benefits from recent architectural improvements (Dynamic Head, EfficientRep backbone, and anchor-free detection) for robust recognition of small and irregular debris. The modular design ensures that improvements from GlobalLocalNet directly translate to higher detection accuracy under complex underwater conditions.

IV. Results and Discussion

A. Data Preparation and Experimental Setup

To train the GL-FPN-SENet for underwater image enhancement, we used the EUVP dataset, which consists of underwater images captured by Remotely Operated Vehicles (ROVs). The dataset is prepared in a paired format, consisting of two groups: low-quality underwater images and corresponding ground-truth reference images. For the application in underwater debris detection, we used the TrashCan 1.0 dataset an instance-segmentation labeled dataset containing 7,212 images of trash observations. The dataset was split into 70% for training, 20% for validation, and 10% for testing. All images were enhanced using the Global-Local Net to improve visibility, making underwater debris more easily detectable.

B. Underwater Image Enhancement

Integrating the Feature Pyramid Network (FPN) into the Local Branch allows the model to combine spatial information across multiple resolutions, restoring both global context and fine-grained details. As shown in Fig. 5, this improves surface textures such as coral patterns, which appear closer to the ground truth. The addition of the Squeeze-and-Excitation (SE) block further emphasizes channels most relevant to structure and color, producing sharper images with more balanced saturation. This is evident in the bottom row of Fig. 5, where small dark objects on sandy backgrounds show clearer contrast. Overall, the model preserves natural color tones more effectively, avoiding flat or greenish hues seen in poor-quality inputs and even in the original Global–Local Net.

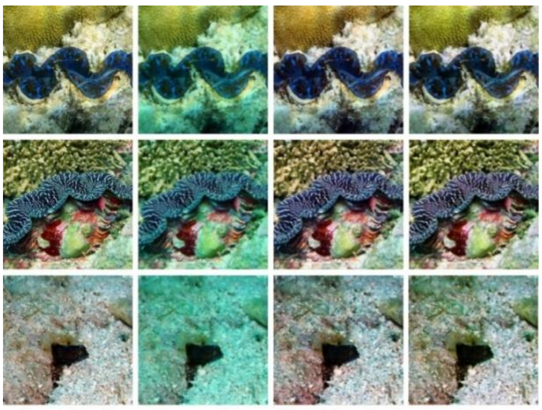

Fig. 5. Visual comparison of underwater image enhancement results.

C. Loss Function

To improve underwater image quality, this study employs a Global–Local Network trained with loss functions that enforce similarity to ground truth. SSIM loss preserves structural details and textures often degraded by turbidity, while Average Color Loss guides global color correction toward realistic tones. To further enhance realism, the model is trained within an adversarial framework, where the generator produces outputs that the discriminator cannot distinguish from real images.

As shown in Table I, the proposed model achieves a higher PSNR 20.732 vs. 19.941, indicating reduced distortion, while maintaining comparable SSIM (0.833 vs. 0.836). These results confirm that the approach improves image quality and visual fidelity without compromising structural consistency.

TABLE I. Average SSIM and PSNR comparisons

Model	PSNR	SSIM
Global-local net	19.941	0.836
Our	20.732	0.833

D. Object Detection Performance with YOLOv12

After the underwater image restoration process, object detection is performed using the YOLOv12 model with the medium configuration and trained for 100 epochs. The evaluation is based on Precision, Recall, and mean Average Precision (mAP), categorized by the types of objects appearing in the images, including marine life, various types of underwater debris, and exploration-related objects such as ROVs. The detection results are presented in Table III, showing a comparison across different object classes.

TABLE II. Average Prediction Scores from YOLOv12m

Model	Precision	Recall	mAP
YOLOv12m	93.22%	90.58%	93.14%

TABLE III. Performance Evaluation of Underwater Accuracy

Class	Precision	Recall	mAP
ROV	93.22%	90.58%	93.14%
Plant	98.1%	91.2%	94.5%
Animal fish	90.7%	87.1%	90.2%
Animal starfish	92.3%	88.9%	91.5%
Animal shells	88.9%	84.6%	88.3%
Animal crab	86.5%	82.4%	86.7%
Animal eel	89.1%	85.2%	88.9%
Animal etc	91.2%	86.9%	90.1%
Trash etc	90.4%	89.6%	91.8%
Trash fabric	92.6%	91.0%	93.2%
Trash fishing gear	95.7%	90.8%	94.1%
Trash metal	96.9%	94.7%	96.3%
Trash paper	98.4%	96.5%	98.1%
Trash plastic	95.2%	93.6%	95.9%
Trash rubber	97.6%	98.2%	98.9%
Trash wood	93.7%	95.1%	95.6%

The results demonstrate that the model can effectively detect underwater objects, particularly those with distinctive visual features such as trash rubber, trash paper, and trash metal, which achieved mAP values exceeding 96%. These results suggest the model's strength in detecting man-made debris with consistent appearance. In contrast, biologically derived objects with greater intra-class variability, such as animal shells and crabs, yielded slightly lower accuracy due to their irregular shapes and textures. Nevertheless, the overall detection performance remains robust and suitable for practical deployment. The average values of all evaluation metrics Precision (93.22%), Recall (90.58%), and mAP (93.14%) are summarized in Table II.

Table III presents the class-wise performance of the proposed underwater debris detection system, evaluated in terms of Precision, Recall, and mAP. The results indicate consistently high accuracy across both animate and inanimate categories, with particularly strong performance in detecting man-made debris.

Among non-debris classes, the model demonstrates robust detection of plants (Precision 98.1%, mAP 94.5%) and ROVs (mAP 93.14%). Detection of animal-related categories shows greater variability. For instance, animal fish and animal crab achieved lower recall scores (87.1% and 82.4%, respectively), suggesting occasional misclassification, possibly due to visual similarity or motion blur in the underwater environment. In contrast, animal etc., a composite class, exhibited balanced performance (Precision 91.2%, Recall 86.9%, mAP 90.1%), indicating that the model generalizes reasonably well for unidentified or less represented species.

Performance in debris classification is notably strong across all types, with trash plastic (mAP 95.9%) and trash wood (mAP 95.6%) achieving top-tier results. The highest recall and precision scores are observed in trash rubber (Precision 97.6%, Recall 98.2%), confirming the model's reliability in detecting high-contrast and structurally distinct materials. Even less frequent or ambiguous categories such as trash etc. and trash fishing gear recorded mAP values above 91%, supporting the model's robustness across diverse waste forms.

Overall, these results validate the effectiveness of the integrated enhancement and detection pipeline, particularly for trash-related classes, where the model consistently achieves mAP values exceeding 93%. The relatively lower scores in certain animal categories highlight potential areas for improvement, such as data augmentation or class-specific fine-tuning.

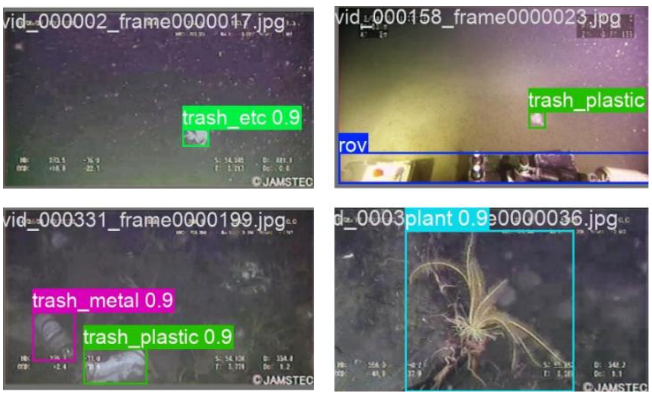

Fig. 6. Detection results from YOLOv12 on underwater ROV images from TrashCan 1.0 dataset.

V. Conclusion

This study presents a high-performance underwater debris detection system using ROV-captured imagery, integrating global–local image enhancement with YOLOv12 in a dual-branch architecture. The Global branch (G-Net) corrects wavelength-dependent color distortions, while the Local branch (L-Net) with FPN and SE modules restores fine details, producing sharper, more natural images for real-time detection.

Experimental results show improved image quality (PSNR rising from 19.941 to 20.732 with SSIM 0.833) and strong detection accuracy. Biological classes such as Plant (mAP 94%) and Starfish (mAP 91%) were detected reliably, though lower performance in Animal crab (86%) and Shells (88%) suggests the need for more diverse training data. In contrast, man-made debris achieved consistently high accuracy (>95% mAP), with trash paper and rubber reaching 98%, confirming the framework's strength in structured object detection.

Beyond accuracy, the framework raises important deployment considerations. Running high-capacity models on AUVs requires hardware optimization, including model compression and energy-efficient inference for embedded GPUs. For large-scale monitoring, cloud-based scalability is essential to process continuous ROV data streams and integrate with multi-modal sources such as satellite and IoT sensors. These factors highlight both the promise and the practical challenges of deploying such systems in real-world marine environments.

In summary, the proposed approach not only advances underwater image enhancement and debris detection but also contributes toward scalable, deployable AI solutions that support marine conservation, biodiversity monitoring, and sustainable ocean governance.

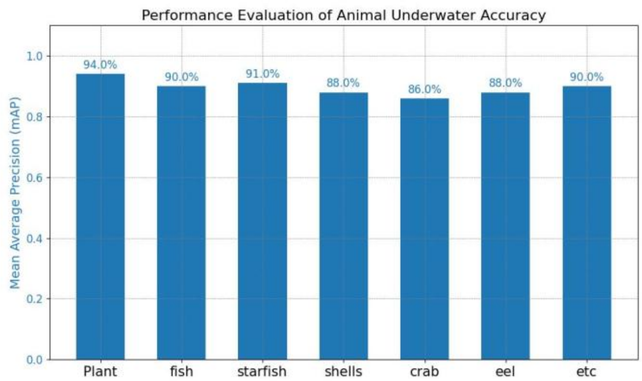

Fig. 7. Bar Chart of Performance Evaluation of Animal Underwater Accuracy.

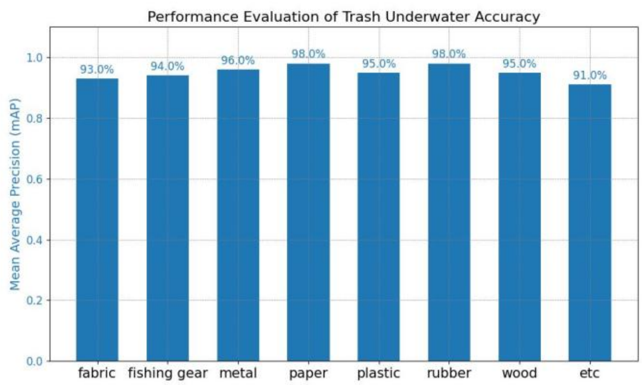

Fig. 8. Bar Chart of Performance Evaluation of Trash Underwater Accuracy.

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